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Techno-economic assessment of a utility-scale wind power plant in Ghana

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ABSTRACT

Ghana's electricity generation mix does not include utility-scale wind power plants to contribute to its power supply. Thus, the country is yet to harness the potential benefits that wind energy could offer, such as diversification of the energy mix, promoting a low-carbon economy, and increasing power generation capacity. This study evaluated wind turbines suitable for developing wind power plants in Adafoah using multicriteria decision-making methods. After that, RETScreen software was employed to perform and evaluate the techno-economic viability of deploying a 10 MW utility-scale wind power plant for electricity generation. The results showed that wind turbines ranging from 1 MW to 1.5 MW are suitable for the study site. Also, developing a 10 MW wind power plant would generate about 93 GWh of electricity annually with a capacity factor of 22%. The findings also reveal that a feed-in tariff (FiT) lower than US$0.15/kWh is not lucrative for wind power development in Adafoah. Furthermore, without financial incentives, the levelized cost of energy (LCOE) from the wind power plant is about US$0.143/kWh. In addition, deploying the wind power plant saves about 33% of carbon dioxide (CO_2) emissions compared to the utility grid. The sensitivity analysis also demonstrates that the project's financial indicators are resilient to capital subsidies, inflation, discount rate, FiT, and wind speed variations. The study's findings suggest that wind energy development in Adafoah and similar locations can improve energy access, reduce greenhouse gas emissions, and promote sustainable development. This study's findings are useful to decision-makers, stakeholders, and investors in developing the wind power sector in Ghana.

1. Introduction

Energy is an indispensable driver for socio-economic growth and development. However, due to climate change, the fuel sources used for electricity generation are a global concern. The rise in electricity consumption, the depletion of fossil fuels, and greenhouse gas (GHG) emissions have sparked investments in renewable energy technologies for electricity generation [53]. Renewable energy is a promising solution for mitigating global GHG emissions and climate change [30]. Wind energy is emerging as one of the world's most promising renewable

Abbreviations: AHP, Analytical hierarchy process; ALCS, Annual lifecycle savings; ANP, Analytic network process; CF, Capacity factor; CO_2, Carbon dioxide; CODAS, Combinative distance-based assessment; COPRAS, Complex proportional assessment; CRITIC, Criteria importance through intercriteria correlation; CSIR, Council for scientific and industrial research; DEA, Data envelopment analysis; EDAS, Evaluation based on distance from average solution; ELECTRE, Elimination and choice translating reality; FiT, Feed-in tariff; GHG, Greenhouse gas; GIS, Geographic information system; GWEC, Global wind energy council; GWh, Gigawatt hour; GWh/yr, Gigawatt hour per year; IEA, International energy agency; IRENA, International Renewable Energy Agency; kg, Kilogram; KWh, Kilowatt hour; LCOE, levelized cost of energy; LWN, Normalise weighted sum of negative distance from average; LWP, Normalise weighted sum of positive distance from average; MCDM, Multicriteria decision-making; MJ, Megajoule; MOORA, multi-objective optimisation on the basis of ratio analysis; MW, Megawatt; NPV, Net present value; O&M, Operation and maintenance; PROMETEE, preference ranking organisation method for enrichment of evaluations; PV, Photovoltaic; SAM, System advisor model; SCY, Specific yield; SF-FAHP, Spherical fuzzy analytic hierarchy process; SPB, Simple payback period; SWARA, Stepwise weighted assessment ratio analysis; T&D, Transmission and distribution; tCO_2, Tonnes of carbon dioxide; TOPSIS, Technique for order preference by similarity to ideal solution; US$, United States dollar; WASPAS, Weighted aggregated sum product assessment; WPD, Wind power density; WPM, Weighted product method; WSM, Weighted sum method; WSN, weighted sum of negative distance from average; WSP, weighted sum of positive distance from average.

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energy sources due to its ability to satisfy growing electricity demand at much lower costs than other renewables [37].

Global wind resources surpass demand, and the installed capacity of wind turbines expanded by more than 20% annually from 2000 to 2019 and is expected to grow by 50% by 2023 [57]. Wind power generated about 273 TWh of electricity in 2021, which is 45% more than what was produced in 2020 [28]. Also, the global wind power capacity has reached 837 GW, saving nearly 1.2 billion tonnes of carbon dioxide (CO_2) emissions annually [25]. Onshore wind is a mature technology, existing in 115 nations globally, whereas offshore wind is in its earliest stages of development, with capacity available in only 19 countries.

As more nations develop or intend to develop their first offshore wind farms, offshore wind is projected to rise in the next few years [28]. The growth of wind power has been driven by declining costs, improved technology, and supportive government policies. The average cost of wind power has fallen dramatically over the past decade, making it increasingly competitive with traditional energy sources such as fossil fuels [27]. In addition, advances in wind turbine technology have improved the efficiency and reliability of wind power systems, increasing their appeal to investors and consumers alike [27].

Approximately 66.4% of Ghana's installed electricity capacity stems from fossil fuels (mainly oil and gas) and 33.9% from large hydropower. Only 0.7% is produced from solar photovoltaics and biogas [49]. The thermal plants' surge in electricity generation could contribute to fuel price growth, fuel supply risks, and carbon emissions. This has subjected Ghana's economy and electricity users to exorbitant and volatile oil-linked electricity prices [13]. Due to limited oil and gas supplies, uncertain water flow, and the need to stabilise economic activities, it is vital to diversify the energy mix.

The demand for electricity in Ghana is rapidly increasing due to the growing population and the country's industrialisation drive. However, most of Ghana's population still lacks access to reliable and affordable electricity, with about 13% lacking access to electricity [20]. This situation has led to exploring renewable energy sources to augment the country's power generation capacity. Wind energy is one of Ghana's most promising and abundant renewable energy resources [18]. In light of this, Ghana passed the Renewable Energy Act, 2011 (Act 832), to promote enabling conditions and accelerate the development and utilisation of renewable resources for generating heat and power [18]. Also, the law requires utilities and bulk consumers to purchase electricity from renewable resources. Moreover, in its recently drafted Renewable Energy Masterplan, Ghana has targeted generating about 275 MW and 325 MW of electricity from utility-scale wind power plants by 2025 and 2030, respectively [19].

Several studies have shown the viability of wind energy in the country, with some areas experiencing high wind speeds suitable for wind power generation. A survey by Ghana Premium Consultant [21] revealed that grid-connected large wind farms with stand-alone wind turbines are excellent for Ghana's coastal regions. The wind available in Ghana is classified under class 4–6 wind resources, mainly along the border with Togo and the highest ridges north-west of Accra [6]. The mountains along the Ghana/Togo border have a wind power density of 600–800 W/m^2 and a wind energy potential of 800 GWh [6]. A recent study by Asare-Addo [7] indicates that Ghana can generate 300 TWh of wind energy annually. In coastal areas, wind speeds between 6 m/s and 7.1 m/s can be harnessed at 50 m above the ground. Despite this potential, the country has no utility-scale wind power plants in the national electricity generation mix.

Multicriteria decision-making (MCDM) methods are popular in energy planning and resource allocation since they enable decision-makers to assess known criteria [49,50]. MCDM helps resolve energy planning's technical, economic, environmental, and socio-political constraints [51]. A well-designed wind farm can harness maximum energy. Therefore, selecting the optimum turbine type for developing a wind power plant is critical. However, numerous realistic and potentially different choice criteria govern the selection process. It requires efficient MCDM methods that effectively integrate the selection criteria into the decision process [62].

The choice of a wind turbine most compatible with the topographical and geographical characteristics of the location where the turbines will be positioned is a crucial step in this development process. Several decision criteria should be taken into account by the turbine selection method [62]. MCDM methods have been used in several notable studies to select optimal wind turbines and sites appropriate for developing wind farms across the globe. For example, J. Wang et al. [77] applied the Dempster-Shafer evidence theory, the technique for order preference by similarity to ideal solution (TOPSIS), and the stepwise weighted assessment ratio analysis (SWARA) method to establish a decision-making model for offshore wind turbine selection. It was observed that these techniques could help reduce the subjectivity of expert judgement and make it easier to come up with a consensus solution for choosing a wind turbine. Nguyen et al. [46] also used the spherical fuzzy analytic hierarchy process (SF-FAHP) and weighted aggregated sum product assessment (WASPAS) to select a suitable wind turbine supplier for a wind energy project in Vietnam. The findings are helpful for investors and decision-makers in other renewable energy projects and also for evaluating and choosing equipment suppliers for wind energy projects. Gil-García et al. [24] integrated a technique for order performance by similarity to ideal solution (TOPSIS), analytical hierarchy process (AHP), and a fuzzy-geographic information system (fuzzy-GIS) to choose the best offshore wind site.

Ma et al. [41] applied the analytic network process (ANP) and the entropy method for selecting wind turbines. Likewise, Deeney et al. [15] utilised the AHP, preference ranking organisation method for enrichment of evaluations (PROMETEE), and elimination and choice translating reality (ELECTRE) to prioritise the best alternative use of a wind turbine blade after its life. Other studies include Supciller & Toprak [72], who employed SWARA, TOPSIS, and evaluation based on distance from average solution (EDAS) for selecting wind turbines. In addition, Beskese et al. [9] chose wind turbines using a hesitant fuzzy AHP-TOPSIS method. The findings indicate that the selection procedure is sensitive to technical and economic criteria but not to changes in vendor-related criteria. Similarly, Rehman & Khan [61] applied a two-tier MCDM based on fuzzy goal programming principles to select optimal wind turbines from 20 different capacities. Sagbansua & Balo [64] and Shirgholami et al. [69] use AHP to choose the best turbine for a wind turbine project. The authors used technical, economic, environmental, and customer attributes as the four main criteria for the decision-making process. Correspondingly, Mostafaeipour et al. [45] used data envelopment analysis (DEA) in selecting the best location for a wind turbine among five cities in the region of East Azerbaijan. Díaz & Soares [17] and Koc et al. [36] combined GIS and AHP to select a suitable site for wind energy projects.

Several authors have also applied different methods or techniques to assess the techno-economic feasibility of wind power development in diverse countries, including Ghana. Some notable studies include Adaramola et al. [1], who applied numerical calculations to assess the wind energy potential in Ghana. Also, Asumadu-Sarkodie & Owusu [8] used RETScreen software to study the potential and economic viability of a 10 MW wind farm in Ghana. The authors discovered that the levelized cost of energy (LCOE) ranged from US$0.081/kWh to US$0.092/kWh, which is lower than the LCOE estimated for Sub-Saharan African countries such as Ghana. In Albania, Serdari et al. [67] used RETScreen to study the feasibility of a 12 MW grid-connected wind power farm. It was estimated that the wind power plant LCOE, simple payback time, and CO_2 emissions reduction are US$0.058/kWh, 5.9 years, and 642,328 tonnes, respectively. Niyomtham et al. [48] employed the WindSim computational fluid dynamics model to propose a 15 MW wind power plant in Thailand. The results show the power plant generates about 41 GWh of electricity annually. This saves about 231 tonnes of CO_2 emissions annually, yielding an LCOE of approximately US$0.093/kWh. Pourasl & Khojastehnezhad [55] also used numerical calculations

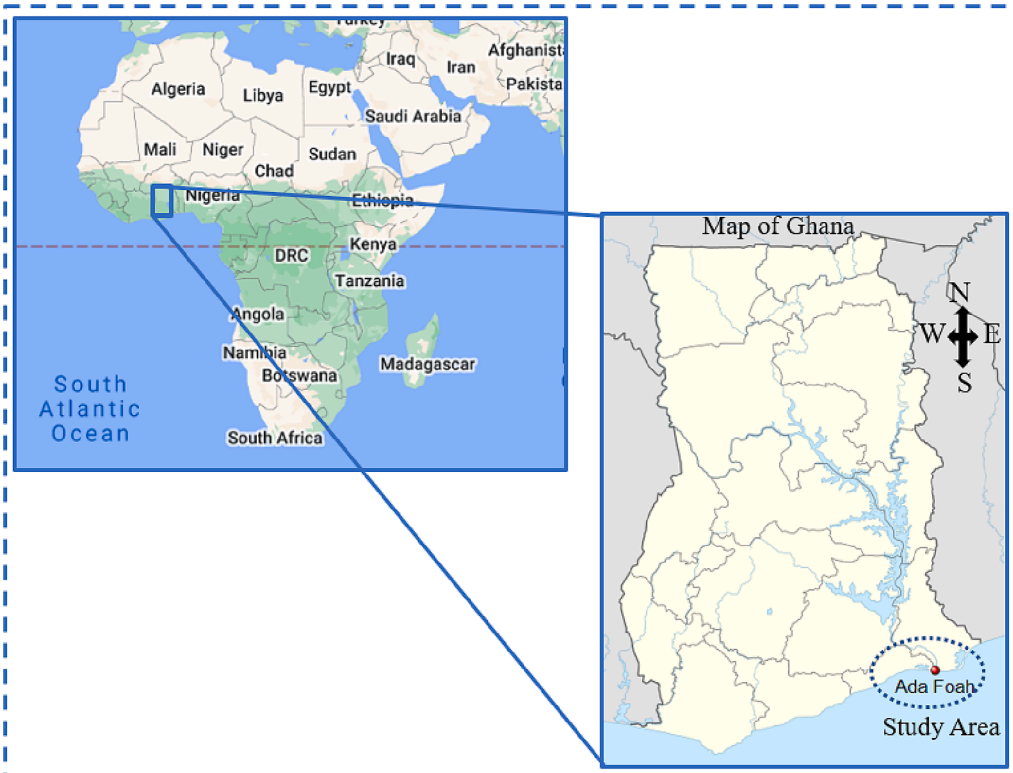

Fig. 1. Location of the study area in the context of Africa and Ghana.

to examine the techno-economic analysis of wind energy potential in Kazakhstan. The authors observed that the annual energy production from the selected turbines ranges from 3.8 GWh/yr to 15.4 GWh/yr. Also, the capacity factors are between 29% and 59%. Likewise, Adeyeye et al. [2] applied numerical computations to analyse wind energy costs for low wind speed areas in Chad, Kenya, Morocco, Namibia, Nigeria, Rwanda, and Tanzania. The study revealed that the Ferris wheel turbine is optimal for increasing Africa's access to affordable, environmentally friendly, and sustainable energy. Correspondingly, Charrouf et al. [12] performed a techno-economic analysis of wind turbines for developing a wind farm in Algeria. The study found that a 1 MW wind turbine is most suitable for the region. Also, the LCOE varies from US$0.0878/kWh to US$0.1142/kWh. Rafique et al. [60] assessed the viability of installing a 100 MW utility-scale wind farm for five different cities in Saudi Arabia and found that all sites are technically and economically viable.

Other studies include Adnan et al. [3], who performed a techno-economic analysis of wind power generation in Pakistan. The findings indicate that using Nordex N90/2500 wind turbines is highly beneficial, generating an LCOE between US$0.056/kWh and US$0.074/kWh and a payback period of about 7 years. Dayal et al. [14] applied wind energy industry-standard software to evaluate the performance of a 10 MW wind farm in Fiji. The findings revealed that the wind farm's mean wind speed, power density, and annual energy generation are below the utility-scale criteria of 6.4 m/s, 300 W/m^2, and 500 MWh/year/turbine at 55 m above ground level. Waewsak et al. [75] conducted a techno-economic assessment of a 20 MW wind power plant in Songkhla Province in southern Thailand. It was seen that the wind power plant could generate about 33 GWh of electricity annually at a 21% capacity factor and LCOE of US$0.087/kWh. Khraiwish Dalabeeh [35] evaluated the capacity factor and predicted LCOE for five locations in Jordan. The results indicate that wind turbines generate electricity at 0.05 to 0.10 US $/kWh, significantly cheaper than diesel fuel in a combined cycle or heavy fuel oil in thermal power plants. Mattar & Guzmán-Ibarra [43] assessed the techno-economic feasibility of offshore wind farm potential in Chile. The results demonstrate that the capacity factors of wind farms range between 40% and 60%. In addition, the LCOE ranges between US $0.10/kWh and US$0.114/kWh, which could be deemed competitive in Chile's current electricity market.

Butt et al. [11] applied the system advisor model (SAM) and RETScreen software to perform a techno-economic and environmental analysis of large-scale wind farms in Pakistan. The authors revealed that the unified power flow controller increased power system handling, making it the best compensating device. The wind farm can also save 1,011,957 tCO_2 over the project's lifespan. Wazir et al. [79] applied the autoregressive integrated moving average model and RETScreen software to perform a techno-economic assessment of a 50 MW wind farm in Gawadar, Pakistan. The findings show that the optimal hub height for wind turbines at the site is 100 m, with a capacity factor of 13.32% and an annual wind electricity generation of 58 GWh. Correspondingly, Niazi [47] investigated the techno-economic viability of a 50 MW wind farm in Gwadar, Pakistan, and found that the best hub elevation for wind turbines is 100 m. This yields 58.31 GWh of energy annually. Similarly, Mehmood et al. (2017) evaluated grid-connected wind turbines along six Pakistani coastlines using RETScreen. The 3 MW wind turbine has a capacity factor of 15–19% and generates 205–280% more electricity than 2 MW wind turbines. Both power-rated wind turbines reduce GHG emissions by 400–2600 tCO_2. In addition, Yaqoob et al. [80] employed RETScreen to assess the feasibility of deploying a 50 MW wind project at four sites in the Sindh province of Pakistan using RETScreen. The results showed that all sites are technically and economically feasible. Still, Hyderabad is the most favourable site, with the highest capacity factor of 41.8% and the lowest simple and equity payback periods of 7.4 years and 4.9 years, respectively, compared to the other sites. Moreover, Khan et al. [34] revealed that installing a 100 MW wind farm in the Gwadar region of Pakistan could produce 298 GWh of electricity annually and reduce 12,648 tonnes of CO_2 emissions. In addition, the wind farm's LCOE has an LCOE of about $US0.09/kWh and a payback period of about 4 years.

The studies mentioned above indicate that several studies on wind power development have been investigated globally. Despite the

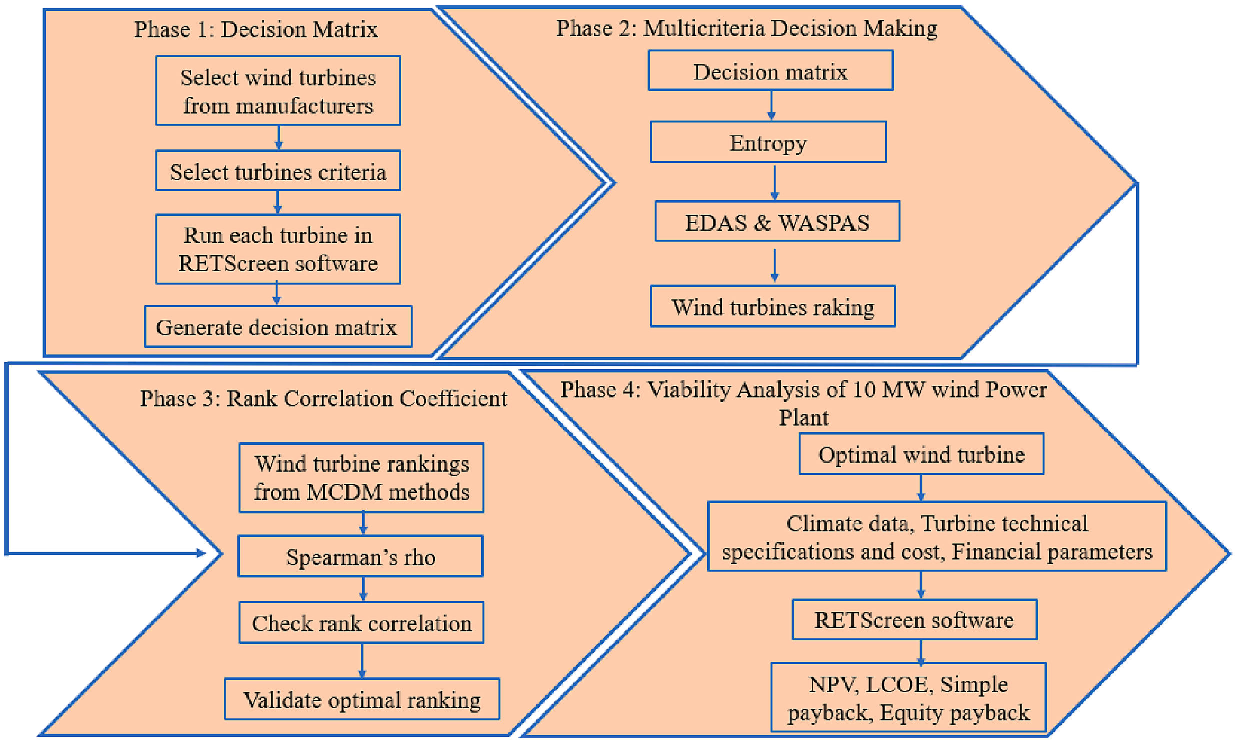

Fig. 2. Flow of study approach with its phases.

potential for wind energy in Ghana, there is a dearth of empirical research on the techno-economic feasibility of utility-scale wind power plants. This research gap limits the development and deployment of wind energy projects in the country, as investors are hesitant to invest in projects without clear information on the project's economic viability. Therefore, there is a need for comprehensive techno-economic assessments of wind power plants in Ghana to support decision-making and provide insights for investors. This study aims to fill this gap by conducting a techno-economic assessment of Ghana's utility-scale wind power plant.

The study seeks to identify the most suitable wind turbine for the site, evaluate the techno-economic feasibility of the wind power project, and estimate its GHG emissions reduction potential. The study's main contributions are outlined as follows: Twenty wind turbines from notable manufacturers ranging from 1 MW to 3 MW suitable for wind power generation are assessed; (2) thirteen (13) salient criteria for decision-making are considered that are limited in other studies; (3) for the first time, MCDM methods have been employed to evaluate wind turbines suitable for Ghana taking into consideration the wind speed conditions; and (4) a viability analysis of a 10 MW utility-scale wind power plant is performed. The findings from this study will provide insights into the techno-economic feasibility of wind power projects in Ghana and support decision-making and policy formulation to encourage private sector investment. The study's results will also contribute to the body of knowledge on wind power in Ghana, promoting the country's transition towards clean and sustainable energy. Further, the study findings can be used as guidance for developing wind power projects in Ghana and other places where the wind speed is low to moderate for the particular wind turbine.

2. Materials and methods

The study approach integrates the following sections to address the study goal: Section 2.1 highlights the study area. Section 2.2 provides a detailed outline of the proposed study approach. Similarly, Section 2.3 highlights the RETScreen expert software. Multicriteria decision-making methods are discussed in Section 2.4. Also, Section 2.4 outlines the rank correlation coefficients approach.

2.1. Brief description of the study area

Adafoah is the location for this study, as shown in Fig. 1. Adafoah is a town on the southeast coast of Ghana, where the Volta River meets the Atlantic Ocean. It is located at a latitude of 5.79° N and a longitude of 0.62° E. The total land area is about 315.5 km^2. Ada Foah is also the capital of the Ada East District (formerly Dangme East District) and the seat of the District Assembly. It is mainly known for its palm-lined beaches and estuary islands. The Adafoah has significant wind speed potential that can be harnessed for electricity generation in Ghana.

2.2. Proposed study approach

The proposed study flow is illustrated in Fig. 2. The study approach comprises four phases as follows: Phase 1 (generating a decision matrix); Phase 2 (MCDM methods); Phase 3 (rank correlation coefficient); and Phase 4 (economic viability of developing the 10 MW wind power plant). All relevant input data is entered into the RETScreen expert software to produce the decision matrix for MCDM analysis. After that, the decision matrix is subjected to entropy to compute criteria weights. This helps determine the statistical significance of each criterion. Entropy is selected because it is an objective method and eliminates the need for an expert. This decreases the vulnerability of decision-making to variations in criteria weights and minimises individual choice. Consequently, it produces accurate results. Besides, entropy is more reliable and resilient than popular objective weighting methods like criteria importance through intercriteria correlation (CRITIC) when integrated into outranking methods [49].

The estimated weights are then integrated into evaluation based on distance from average solution (EDAS) and weighted aggregated sum product assessment (WASPAS) to rank the wind turbines. The Spearman rank correlation coefficient test is applied to ascertain the correlation between the rankings obtained from the MCDM methods. This would

Table 1
Key input parameters adopted in the RETScreen wind energy project model.

Parameter	Value	Remarks/source
Wind speed (m/s)	5.3	Average annual wind speed for Adafoah at 12 m height [1].
Wind shear exponent	0.14	Low exponents indicate smooth terrain, while large exponents indicate obstructions. When site characteristics are unknown, 0.14 is an acceptable approximation [8].
Array losses (%)	7	Losses caused by turbine spacing, orientation, site characteristics, and topography. Data obtained from [8]
Airfoil soiling losses (%)	2	Losses due to blade soiling from bugs and/or ice build-up [26].
Miscellaneous (%)	2	Losses in energy production due to starts and stops, off-yaw operation, high wind, and wind gust cut-outs. Data assumed from [26].
Downtime losses (%)	10	Losses from scheduled maintenance, wind turbine failure, station outage, and utility outage. Data retrieved from [8].
Shape factor	2	Typically, the shape factor falls between 1 and 3. Assumed value deduced from [1].
Availability (%)	96	Average value deduced from [8,70].
Fuel cost escalation rate (%)	2	Typical values range from 0 to 5%, with 2 to 3% being the most common.
Inflation rate (%)	10	Current inflation rate in Ghana at the study time.
Discount rate (%)	10	Typical discount rate for similar projects in developing countries [52,71].
Capital cost (US$/kW)	1,409	This cost covers equipment, the balance of the system, and installation. Average value obtained from [29,70]
Miscellaneous (US$)	200,000	The cost of developing a substation and transmission line. Assummed from [8].
Electricity export rate	0.166	Approved feed-in tariff for wind power plants in Ghana (Public Utilities Regulatory [58]
Transmission losses	4.5	Typical power loss in Ghana's distribution network [20]
Operation and Maintenance (US $/kW)	44.5	Average value deduced from [29,70]
GHG emission factor (tCO_2/MWh)	0.36	Ghana's grid emission factor for wind and solar projects [20]
Project lifespan (years)	20	Typical wind power projects are between 20 and 25 years

help determine the optimal wind turbine suitable for developing a wind power plant for the study site. After that, RETScreen software is further employed to investigate the techno-economic viability of electricity generation from the optimal wind turbine.

2.3. RETScreen wind energy project model

RETScreen Expert software is sustainable energy management software for analysing energy efficiency, renewable energy, cogeneration plant feasibility, and performance. The software empowers experts and decision-makers to quickly define, evaluate, and maximise future renewable energy projects' technological and financial feasibility. It is commonly used in energy systems for benchmarking, feasibility, and performance analysis [56]. The wind energy project model in RETScreen allows users to perform wind turbine design and analysis to optimise supply. The software retrieves climatic data, including temperature, relative humidity, precipitation, atmospheric pressure, and solar irradiation, based on the latitude and longitude of the study location to perform the required analysis. Table 1 summarises the most important input variables used in the wind energy project model in RETScreen.

2.3.1. Wind speed distribution

RETScreen computes the wind speed distribution as a Weibull probability density function as follows [63]:

$$p(x) = \left(\frac{k}{C}\right)\left(\frac{x}{C}\right)^{k-1}\exp\left[-\left(\frac{x}{C}\right)^{k}\right] \tag{1}$$

where, p(x) is the probability, k is the shape factor, x is the wind speed, and c is the scale factor. Eq. (1) is valid when $k > 1,\ v \geq 0,$ and $c > 0$. The scale factor is further expressed as follows:

$$C = \frac{\overline{x}}{\Gamma\left(1+\frac{1}{k}\right)} \tag{2}$$

Where, $\overline{x}$ is the average wind speed, and Γ is the gamma function. RETScreen computes the wind speed distribution from the wind power density. Thus, the wind power density and the average wind speed are given as follows [63]:

$$WPD = \sum_{X=0}^{X=25}\frac{1}{2}\rho x^3 p(x) \tag{3}$$

$$\overline{x} = \sum_{X=0}^{X=25}\frac{1}{2}xp(x) \tag{4}$$

Where, ρ is the air density, WPD is the wind power density and $\overline{x}$ is average wind speed.

2.3.2. Energy curve

Energy curve data shows how much energy a wind turbine generates at different annual average wind speeds. RETScreen specifies the energy curve over 3–15 m/s annual average wind speed. Eq. (5) is applied to determine the energy curve:

$$E_{\overline{v}} = 8760\sum_{x=0}^{x=25}P_x p(x) \tag{5}$$

Where P_x is the turbine power, and $p(x)$ is the Weibull probability density function.

2.3.3. Unadjusted energy production

Unadjusted turbine energy is produced at a standard temperature and pressure. The computation uses the site's average hub-height wind speed. Due to wind shear, hub-height wind speed is frequently higher than anemometer height. Eq. (6) is used to calculate the average wind speed at hub height [49]:

$$\frac{\overline{x}}{\overline{x}_0} = \left(\frac{H}{H_0}\right)^{\alpha} \tag{6}$$

Where, $\overline{x}$ is the average wind speed at hub height H, $\overline{x}_0$ is the wind speed at an anemometer height H_0, and α is the wind shear exponent. Interpolating the energy curve from Section 2.3.3 at the hub height yields the unadjusted energy production.

2.3.4. Gross energy production

Gross energy production is the total energy the wind turbine produces annually without losses using the site's wind speed, atmospheric pressure, and temperature conditions. Eq. (7) is applied to estimate the gross energy production:

$$E_G = E_U\left(\frac{K}{K_O}\right)\left(\frac{M_O}{M}\right) \tag{7}$$

Where, E_G is the gross energy production, E_U is the unadjusted energy production, K is the site's annual average atmospheric pressure, K_0 is the standard atmospheric pressure of 103.3 kPa, M is the site's annual average absolute temperature, and M_0 is the standard absolute temperature of 288.1 K.

2.3.5. Energy collected

RETScreen model determines renewable energy collected (equal to the net amount of energy produced by the wind turbines) as follows:

$$E_C = E_G(1-\gamma_a)(1-\gamma_{asi})(1-\gamma_d)(1-\gamma_m) \tag{8}$$

Where, E_C is the renewable energy collected, γ_a is the array losses, γ_{asi} is the losses due to airfoil soiling, γ_d is the losses caused by downtime, and γ_m is the miscellaneous losses.

2.3.6. Specific yield

The model estimates the wind power plant specific yield (SCY) using Eq. (9):

$$SCY = \frac{E_C}{(TN)\,(SW)} \tag{9}$$

Where, TN is the total number of turbines, and SW is the turbine rotor swept area.

2.3.7. Capacity factor

The wind plant capacity factor (CF) is the ratio of the annual energy generated by the wind power plant to its rated power capacity. The CF is computed using Eq. (10):

$$CF = \left(\frac{E_C}{WTp_c OP_{hr}}\right) \times 100 \tag{10}$$

where E_C is the renewable energy collected (kWh), WTP_c is the wind power plant capacity (kW), and OP_{hr} is the wind turbine operational hours per year.

2.3.8. Economic feasibility indicators

The 10 MW wind power plant's economic viability is assessed based on the following indicators: net present value (NPV), LCOE, simple payback, equity payback, and annual lifecycle savings. NPV is the difference between the present value of the project's cash inflows (benefits) and the present value of the project's cash outflows (cost). The NPV is computed using Eq. (11) [26,56]. Positive NPV values (NPV > 0) indicate a potentially feasible project.

$$NPV = \sum_{n=0}^{N} \frac{AC_n}{(1+r)^n} \tag{11}$$

Where, AC_n is the after-tax cash flow in n number of years, r is the discount rate, and N is the project life in years,

The simple payback is the time it requires a cash flow to equal the investment. It determines the time required to recoup the investment. The project would be financially feasible when the simple payback is less than the project's economic lifespan [48]. The model computes the simple payback using Eq. (12):

$$\text{Simple payback} = \frac{TC - InG}{(E_{sav} + C_{sav}) - (OM_{cost} + F_{cost})} \tag{12}$$

Where, TC is the project's initial capital cost, InG is the available financial incentives, E_{sav} is the annual energy savings, C_{sav} is the annual capacity savings, OM_{cost} is the annual operation and maintenance cost and F_{cost} is the annual fuel or electricity cost.

The annual lifecycle savings (ALCS) is the levelized nominal annual savings with the same lifespan and NPV as the project. It is computed by using Eq. (13):

$$ALCS = \frac{NPV}{\frac{1}{r}\left(1 - \frac{1}{(1+r)^N}\right)} \tag{13}$$

The levelized cost of energy (LCOE) is computed to examine the wind power plant's energy generation cost. The LCOE, as given in Eq. (14), represents the electricity export rate required to have an NPV = 0, that is, the average energy cost produced throughout the project's lifetime [56].

$$LCOE = \frac{\sum costs}{\text{annual energy production (MWh)} \times \text{project lifetime}} = \frac{\sum \frac{costs_n}{(1+i)^n}}{\sum \frac{energy_n}{(1+i)^n}} \tag{14}$$

Table 2
Criteria attributes and their respective impact on the decision-making processes.

Criteria	Index	Attribute Type	Significance
Rated capacity (kW)	C_{01}	positive (+)	This defines the character of a specific wind turbine and is furnished by the original equipment manufacturer. The rated capacity is reached at the rated wind speed.
Capacity factor (%)	C_{02}	positive (+)	It is the ratio of the annual energy generated by the wind power plant to its rated power capacity. Typical values for wind plant capacity factors range from 20 to 40%.
Gross energy production (kWh/yr)	C_{03}	positive (+)	It refers to the total energy the wind turbine produces annually without losses under the site's conditions.
Specific yield (kWh/m^2)	C_{04}	positive (+)	It compares the performance of a wind turbine to the local wind regime. The specific yield is calculated by dividing the energy output by the rotor's swept area.
Energy available for grid export (kWh/yr)	C_{05}	positive (+)	This is the electricity the power system sends to the grid after distribution losses.
Hub height (m)	C_{06}	negative (−)	This refers to the height at which the wind turbine rotor is mounted. Increasing the hub height should be considered to enhance the project's energy output when possible.
Rotor diameter (m)	C_{07}	negative (−)	This defines the diameter of the circle formed by the rotation of the wind turbine blades. The rotor diameter of the commercially available turbines typically ranges from 7 to 120 m or more.
Swept area (m^2)	C_{08}	negative (−)	It refers to the area a rotor covers perpendicular to the wind direction during one complete rotation. Wind turbine power and energy output depending on the rotor-swept area.
Cut-in speed (m/s)	C_{09}	negative (−)	This denotes the minimum wind speed the turbine requires to start generating energy. The turbine's original equipment manufacturers specify it.
Rated wind speed (m/s)	C_{10}	negative (−)	This denotes the wind speed at which the electrical generator reaches its rated power (maximum output power). When designing the blades of a wind turbine, the rated wind speed is essential.
Capital Cost (US $/kW)	C_{11}	negative (−)	This initial cost includes equipment and installation costs for the wind farm's deployment.
CO_2 emissions (tCO_2/yr)	C_{12}	negative (−)	It signifies the total CO_2 emitted per year from the proposed system.
Operating and maintenance cost (US$/kW/yr)	C_{13}	negative (−)	This is the total cost of running and maintaining the system over the project's life.

2.3.9. Greenhouse gas emissions and savings

The model computes GHG emissions generated by the wind power plant as follows:

$$GHG_{emissions} = E_w * E_{grid} * \beta_p \quad (15) \tag{15}$$

Where, $GHG_{emissions}$ is the greenhouse gas emissions produced by the wind power plant (tCO_2), E_w is the wind turbine greenhouse gas emission factor (tCO_2/kWh), E_{grid} is the electricity available for grid export (kWh) and β_p electricity transmission and distribution (T&D) losses. In addition, the annual greenhouse gas emission reduction is estimated using Eq. (16):

$$GHG_{red} = (E_g - E_w)E_{grid}(1 - \beta_p) \tag{16}$$

Where,GHG_{red} is the annual GHG emissions reduction (tCO_2) E_g is the grid GHG emission factor (tCO_2/kWh), and E_w is the wind turbine's GHG emission factor (tCO_2/kWh).

2.4. Multicriteria decision making methods

MCDM methods are sophisticated decision-making techniques designed to address challenges and systems with multiple criteria. The methods are primarily used to compute weights and rank alternatives and options [81]. Table 2 lists the criteria attributes and their respective impacts on decision-making processes.

2.4.1. Entropy

Entropy is a newly developed MCDM model for weight determination that employs the initial decision matrix to compute criteria weights. Entropy eliminates the subjectivity of expert judgements, resulting in a more stable and accurate performance evaluation [39]. The method integrates Eq. (17) to Eq. (19) to compute criteria weights [32,42,81]:

Step 1: Normalised the decision matrix using Eq. (17):

$$S_{ij} = Y_{ij} / \sum\nolimits_{i=1}^{n} Y_{ij} \; i = 1, 2, ?, n \tag{17}$$

where, S_{ij} represents the normalised value of the ith alternative for the jth criterion, and Y_{ij} represents the decision matrix values of the ith alternative for the jth criterion.

Step 2: Compute the criteria entropy values as follows:

$$N_j = -R \sum_{i=1}^{n} S_{ij} \log S_{ij}, \; j = 1, 2, ?., n \tag{18}$$

Where, $R = \frac{1}{\log_a}$ is a constant that ensures $0 \le N_j \le 1$, N_j is the entropy value with respect to criterion, and a is the number of wind turbines.

Step 3: Estimate criterion objective weight using Eq. (19):

$$W_j = 1 - N_j / \sum\nolimits_{i=1}^{n} 1 - N_j, \; j = 1, 2, \cdots, n \tag{19}$$

where, W_j is the weight for each criterion C_j.

2.4.2. Evaluation based on distance from average solution (EDAS)

EDAS is among the newly developed and utilised ranking methods in integrated decision-making models. EDAS uses a different normalisation type for each criterion's average solutions [23,81]. Also, this technique is helpful when there are conflicting criteria. EDAS calculates wind turbine performance using Eqs. (19)–(26) [51,52]:

Step 1: Find criterion average solution $((M_j))$ using Eq. (20):

$$M_j = \frac{\sum_{i=1}^{n} Y_{ij}}{c} \tag{20}$$

Where, Y_{ij} are the elements of the decision matrix and c is the number of criteria.

Step 2: Compute the positive distance from average (PMij), where Eq. (21a) is applied if criterion type is positive and Eq. (21b) is used if criterion type is negative:

$$PM_{ij} = \frac{\max(0, (Y_{ij} - M_j))}{M_j} \tag{21a}$$

$$PM_{ij} = \frac{\max(0, (M_j - Y_{ij}))}{M_j} \tag{21b}$$

Step 3: Estimate the negative distance from average (NMij) where Eq. (22a) is applied if criterion type is positive and Eq. (22b) is used if criterion type is non-negative.

$$NM_{ij} = \frac{\max(0, (M_j - Y_j))}{M_j} \tag{22a}$$

$$NM_{ij} = \frac{\max(0, (Y_{ij} - M_j))}{M_j} \tag{22b}$$

Step 4: Calculate the weighted sum of PM_{ij} and NM_{ij} for the wind turbines using Eq. (23) and Eq. (24), respectively:

$$WSP_i = \sum_{j=1}^{m} w_j PM_{ij} \tag{23}$$

$$WSN_i = \sum_{j=1}^{m} w_j NM_{ij} \tag{24}$$

Where, WSP_i is the weighted sum of positive distance from average and WSN_i is the weighted sum of negative distance from average.

Step 6: Normalise WSP_i and WSN_i values as follows:

$$LWP_i = \frac{PM_i}{\max_i(PM)} \tag{25}$$

$$LWN_i = 1 - \frac{NM_i}{\max_i(NM_I)} \tag{26}$$

Where, LWPi and LWNi are the normalised values for WSPi and WSNi, respectively.

Step 7: Compute the performance score (Pi) for each wind turbine based on Equation (26):

$$P_i = \frac{1}{2}(LWP_i + LWN_i) \tag{27}$$

2.4.3. Weighted aggregated sum product assessment (WASPAS)

WASPAS combines the weighted sum method (WSM) and the weighted product method (WPM). The WPM and WSM are extensively employed to select the optimal solution for a decision-making problem [73]. In addition, WASPAS is among the most recent methods that can improve the accuracy of selecting the best option. Besides, the WASPAS method is more precise than the WPM and WSM methods [76]. The WASPAS method combined the following steps to rank the wind turbines [38,73,76]:

Step 1: Construct a decision matrix as follows:

$$X = [x_{ij}]_{m \times n} \tag{28}$$

Where, x_{ij} is the performance of the ith alternative with respect to the jth criterion, m is the number of alternatives, and n is the number of criterion.

Step 2: Normalised the decision. Eq. (27) a is applied to positive attributes. While Eq. (29b) is used for negative attributes:

$$\overline{X}_{ij} = \frac{x_{ij}}{\max_i x_{ij}} \tag{29a}$$

$$\overline{X}_{ij} = \frac{\min_i x_{ij}}{x_{ij}} \tag{29b}$$

Step 3: Estimate the significance of the ith alternative as follows:

Table 3
Estimated criterion weights.

Sn	Criteria	Index	N	$1\text{-}N_j$	W_j
1	Rated capacity	C_{01}	0.9782	0.0218	0.1061
2	Capacity factor	C_{02}	0.9959	0.0041	0.0198
3	Gross energy production	C_{03}	0.9703	0.0297	0.1445
4	Specific yield	C_{04}	0.9971	0.0029	0.0140
5	Energy available for grid export	C_{05}	0.9708	0.0292	0.1420
6	Hub height	C_{06}	0.9951	0.0049	0.0238
7	Rotor diameter	C_{07}	0.9928	0.0072	0.0348
8	Swept area	C_{08}	0.9723	0.0277	0.1344
9	Cut-in speed	C_{09}	0.9967	0.0033	0.0161
10	Rated wind speed	C_{10}	0.9973	0.0027	0.0130
11	Capital cost	C_{11}	0.9782	0.0218	0.1061
12	CO_2 emissions	C_{12}	0.9713	0.0287	0.1393
13	Operating and maintenance cost	C_{13}	0.9782	0.0218	0.1061

$$Q_1^{(WSM)} = \sum_{j=1}^{n} \overline{X}_{ij} W_j \quad (30)$$

Where, is W_j weight (relative importance) of the *j*th criterion

Step 4: Compute the total importance of ith alternative as follows:

$$Q_2^{(WPM)} = \prod_{j=1}^{n} \left(\overline{X}_{ij}\right)^{W_j} \quad (31)$$

Step 5: Determine the wind turbines performance score (Q_i) using Eq. (32):

$$Q_i = 0.5Q_1^{(WSM)} + 0.5Q_2^{(WPM)} \quad (32)$$

2.5. Spearman's rho (ρ) correlation coefficient

Spearman's correlation coefficient is a non-parametric statistic for measuring rank correlation. Eq. (33) is applied to analyse the correlations between the wind turbines:

$$\rho = 1 - \frac{6\sum_{i=1}^{n} d_i^2}{n(n^2 - 1)} \quad (33)$$

where, ρ is the Spearman's rank correlation, d_i is the difference between the two ranks of each wind turbine, and n is the total number of wind turbines under assessment.

3. Results and discussions

3.1. Multicriteria decision making for the selection of optimal wind turbine

Table A1 presents the extracted decision matrix for MCDM analysis. Significant observations from Table A1 reveal that it is difficult to ascertain the optimum wind turbine straightforwardly since the decision matrix has significant variations in criteria values. However, applying the proposed MCDM methods to the decision matrix would resolve this situation and determine the wind turbine performance. The turbine criteria weights play a significant role in determining the best wind turbines. The criteria weights greatly influence decision-making because the MCDM methods depend on weights to assess the wind turbine's performance. Table 3 shows the weights computed for each criterion. The total criteria weight equals one. The results show that the criteria weights are between 1% and 14%. It can be gleaned that criteria including C_{03} (gross energy production), C_{05} (energy available for export), C_{08} (swept area), and C_{12} (CO_2 emissions) have the highest criteria weights of about 13% to 14%. The implication is that these criteria might strongly influence the final optimal wind turbine selection. Next are C_{01} (rated capacity), C_{11} (capital cost), and C_{13} (operating and maintenance cost), which weigh about 11%. The remaining criteria have the lowest weights, below 5%, and would have a lower impact on the final turbine rankings. Though significant variations exist between the criteria weights, the variation could be considered minimal. For example, the difference between the highest and intermediate weights is only 24%. Also, the difference between the intermediate and lowest weights is only 167%. Consequently, the weight difference between the highest and lowest weights is about 173%. As mentioned above, the weights are merged into EDAS and WASPAS to select the optimal wind turbine suitable for Adafoah.

Table 4
Wind Turbines ranking based on EDAS approach.

Index	Turbine name	WSP_i	WSN_i	LWP_i	LWN_i	P_i	Rank
Turbine 1	SL1500/ 70–65 m	0.161	0.133	0.523	0.650	0.587	4
Turbine 2	SL3000/ 100–110 m	0.231	0.260	0.752	0.315	0.534	11
Turbine 3	AW-70/1500 Class I – 80 m	0.149	0.124	0.486	0.673	0.579	5
Turbine 4	DEWIND 62–68.5 m	0.257	0.203	0.838	0.467	0.652	2
Turbine 5	ENERCON – 82 E2 2 MW – 98 m	0.042	0.019	0.137	0.951	0.544	10
Turbine 6	NORDEX S77 – 90 m	0.116	0.101	0.377	0.734	0.556	8
Turbine 7	G80-2 MW – 60 m	0.056	0.062	0.182	0.836	0.509	15
Turbine 8	G80 RCC – 78 m	0.062	0.061	0.201	0.840	0.521	13
Turbine 9	VESTAS V90- 3.0 MW – 90 m	0.133	0.172	0.435	0.548	0.491	19
Turbine 10	VESTAS V90- 1.8 MW – 80 m	0.024	0.034	0.077	0.911	0.494	17
Turbine 11	LTW90-2 MW	0.066	0.043	0.215	0.888	0.551	9
Turbine 12	LTW90 1 MW	0.152	0.143	0.495	0.623	0.559	7
Turbine 13	A-1000/S – 70	0.288	0.236	0.938	0.378	0.658	1
Turbine 14	A-1500-80	0.118	0.099	0.383	0.740	0.562	6
Turbine 15	FL2500/ 90–85 m	0.071	0.098	0.230	0.742	0.486	20
Turbine 16	FL3000 – 100 m	0.307	0.380	1.000	0.000	0.500	16
Turbine 17	HSW 1,000/ 57–55 m	0.280	0.235	0.912	0.381	0.646	3
Turbine 18	W2E-100/ 2.5–85 – 100 m	0.125	0.159	0.407	0.581	0.494	18
Turbine 19	WWD-3 - D90 – 100 m	0.169	0.188	0.551	0.505	0.528	12
Turbine 20	ECO 122–139 m	0.263	0.316	0.857	0.170	0.513	14

Table 4 presents the performance of the wind turbines based on the EDAS approach. The WSP_i and WSNi values were calculated using Eq. (23) and Eq. (24). Consequently, the LWP_i and LWN_i values were estimated based on Eq. (25) and Eq. (26), respectively. Thereafter, the LWP_i and LWN_i were further integrated into Eq. (27) to compute the wind turbine performance score (Pi). EDAS ranks the wind turbines with the highest P_i as optimal. The results show that the wind turbines Pi are nearly close to each other, with a difference between the lowest and highest score being about 30%. This demonstrates a strong correlation between wind turbine performance. However, the results show that the wind turbines with the best rankings are turbine 13 (first position), followed by turbine 4, turbine 17 (third position), turbine 1 (fourth position), and turbine 3 (fifth position). In contrast, the wind turbines with the worst rankings are turbine 16 (sixteenth position), turbine 10 (seventeenth position), turbine 18 (eighteenth position), turbine 9 (nineteenth position), and turbine 15 (twentieth position). This result reveals that wind turbines with a lower-rated capacity of 1 MW to 1.5 MW are appropriate for the study site. That cannot be said for the worst

Table 5
Wind Turbines ranking based on WASPAS approach.

Index	Turbine name	Q_1	Q_2	S_i	Rank
Turbine 1	SL1500/70–65 m	0.5525	0.5292	0.5408	7
Turbine 2	SL3000/100–110 m	0.5798	0.5008	0.5403	8
Turbine 3	AW-70/1500 Class I – 80 m	0.5468	0.5291	0.5379	9
Turbine 4	DEWIND 62–68.5 m	0.6337	0.5505	0.5921	3
Turbine 5	ENERCON – 82 E2 2 MW – 98 m	0.5413	0.5270	0.5341	11
Turbine 6	NORDEX S77 – 90 m	0.5343	0.5226	0.5285	14
Turbine 7	G80-2 MW – 60 m	0.5219	0.5111	0.5165	18
Turbine 8	G80 RCC – 78 m	0.5215	0.5149	0.5182	17
Turbine 9	VESTAS V90-3.0 MW – 90 m	0.5447	0.4987	0.5217	15
Turbine 10	VESTAS V90-1.8 MW – 80 m	0.5207	0.5096	0.5151	19
Turbine 11	LTW90-2 MW	0.5473	0.5251	0.5362	10
Turbine 12	LTW90 1 MW	0.5752	0.5284	0.5518	4
Turbine 13	A-1000/S – 70	0.6766	0.5502	0.6134	1
Turbine 14	A-1500–80	0.5370	0.5249	0.5310	13
Turbine 15	FL2500/90–85 m	0.5278	0.5019	0.5148	20
Turbine 16	FL3000 – 100 m	0.6028	0.4868	0.5448	5
Turbine 17	HSW 1,000/57–55 m	0.6569	0.5403	0.5986	2
Turbine 18	W2E-100/2.5–85 – 100 m	0.5416	0.4991	0.5204	16
Turbine 19	WWD-3 - D90 – 100 m	0.5615	0.5065	0.5340	12
Turbine 20	ECO 122–139 m	0.5897	0.4941	0.5419	6

turbine rankings attributed to the highest-rated capacities between 2.5 MW and 3 MW. This might be due to the available wind speed at the study site.

The wind turbine performance based on the WASPAS approach is given in Table 5. The wind turbines' Q_1 and Q_2 values were computed using Eqs. (30) and (31), respectively. Eq. (32) was used to estimate the S_i values. The results show that the performance scores based on the WASPAS approach exhibit a trend similar to EDAS. However, the difference between the lowest and highest score is lower, about 17%. It is observed that there is a slight deviation between the best and worst-performing turbines compared to EDAS. In view of this, four other methods, including complex proportional assessment (COPRAS) [5,16], the technique of order preference similarity to the ideal solution (TOPSIS) [16], combinative distance-based assessment (CODAS) [65], and multi-objective optimisation on the basis of ratio analysis (MOORA) [49,66], are further tested on the decision matrix to compare the rankings obtained from WASPAS and EDAS as shown in Fig. 3. The implication is that using several MCDM methods on the same decision matrix greatly enhances each method's benefits since each method has its own constraints. Moreover, numerous MCDM methods produce more reliable and plausible findings than a single method [49]. It can be seen in Fig. 3 that the positions of the turbines vary with different MCDM methods. For instance, turbine 13 ranks first in the remaining approaches except in MOORA. Likewise, turbine 17 ranks second in TOPSIS, WASPAS, and COPRAS but third in EDAS, MOORA, and CODAS. A similar trend is seen with turbine 4, which ranks first in MOORA, second in EDAS, third in TOPSIS, WASPAS, and COPRAS, and fifth in CODAS. These variations might be attributed to the advantages and limitations of each MCDM method. This result indicates that the final choice is uncertain due to variations in the methods' rankings, which requires further investigation. Table 6 shows the Spearman rank correlation between the ranks obtained from the MCDM. Correlation

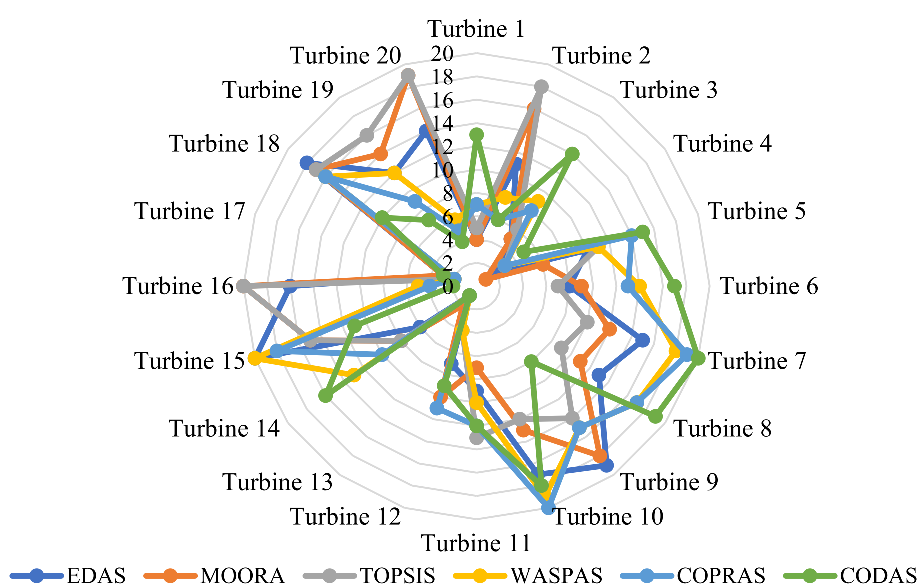

Fig. 3. Comparison of wind turbines rankings to other MCDM methods.

Table 6
Rank correlation coefficient test on wind turbines rankings.

MCDM methods	COPRAS	WASPAS	TOPSIS	MOORA	EDAS	CODAS	Mean ρ
COPRAS	1	0.929	0.269	0.35	0.68	0.785	0.669
WASPAS	0.929	1	0.391	0.432	0.725	0.74	0.703
TOPSIS	0.269	0.391	1	0.893	0.806	−0.09	0.545
MOORA	0.35	0.432	0.893	1	0.878	−0.075	0.580
EDAS	0.68	0.725	0.806	0.878	1	0.2	0.715
CODAS	0.785	0.74	−0.09	0.075	0.2	1	0.452

Table 7
Summary of wind power plant technical, economic, and emission output.

Parameter	Unit	Value
Unadjusted energy production	MWh/yr	23,490.70
Gross energy production	MWh/yr	22,270.90
Electricity export to the grid	MWh/yr	19,103.03
Pressure coefficient	–	0.9859
Temperature coefficient	–	0.9617
Specific yield	kWh/m^2	834.19
NPV	US$	3,807,394.89
LCOE	US$/kWh	0.143
Simple payback	Years	5.2
Annual lifecycle savings	US$/yr	447,215.17
Equity payback	Years	5.6
Annual GHG emissions	tCO_2	6,048
Annual GHG emissions reduction	tCO_2	3,038

coefficients have values ranging from −1 to +1. A positive correlation means both variables are increasing. A negative correlation implies that as one variable's rank increases, the other variable's rank decreases. Table 6 demonstrates a strong correlation between the MCDM method rankings, with a total mean value of the correlation coefficient $\rho = 0.606$. EDAS and WASPAS have a strong positive correlation of $\rho = 0.725$. This is significantly higher than the average correlation coefficient value of $\rho = 0.606$. It can also be observed that EDAS exhibits the highest positive correlation, $\rho = 0.878$ when compared with MOORA. Similarly, when compared to COPRAS, WASPAS has the highest correlation value, $\rho = 0.929$. Furthermore, it can be deduced from the results that the mean ρ values of EDAS and WASPAS are far higher than the other methods. The implication is that the rankings obtained from these MCDMs are optimal compared to the others. Moreover, the EDAS ρ-value is about 2% higher than WASPAS. This suggests that the wind turbine rankings from the EDAS rankings could be the best for determining the optimal wind turbines. Based on these findings, it can be said that Turbine 1 is optimal for developing a wind power plant in Adafoah based on the wind speed and other assumptions applied in this study.

3.2. *Technical, economic, and emission analysis of the proposed 10 MW wind power plant*

3.2.1. *Electricity generation and capacity factor*

Table 7 presents the technical, economic, and greenhouse gas emissions output for the proposed 10 MW wind power plant. The results show that the unadjusted energy production from the wind power plant is about 23,490.69 MWh/yr. This energy generation is reduced to 22,270.90 MWh/yr due to the pressure adjustment and temperature adjustment coefficients. The pressure coefficient determined falls between the recommended range of 0.59 and 1.02. Likewise, the temperature adjustment coefficient (0.9617) is nearly comparable to a typical coefficient between 0.98 and 1.15. Similarly, the loss coefficient for one wind turbine, which integrates all the loss factors, is about 0.8574%, far greater than 0.75 (for a poorly designed project). In addition, the specific yield per turbine, which is frequently used to assess and contrast the performance of a wind turbine, is nearly 834.1937 kWh/m^2. This result is within the typical range of 150 to 1500 kWh/m^2, which is usually accepted for a good wind project.

The annual electricity available for grid export is about 19,103.03 MWh when the power plant availability is assumed to be 96%, and the losses (array, airfoil, and miscellaneous) are taken as 11% (Table 7). The annual electricity available for grid export was observed to depend on wind speed, pressure coefficient, temperature coefficient, and losses. Hence, changing wind speeds and reductions in losses could affect the levelized cost of energy. The wind power plant's monthly electricity generation and capacity factor for the site are shown in Fig. 4. It can be gleaned from Fig. 4 that electricity generation varies throughout the year. However, the difference between the highest and lowest months of electricity generation is about 12%. This means that the monthly variation in electricity is minimal. The capacity factor is mainly influenced by the wind farm's wind resource and the turbine's balance of systems technologies. The capacity factor increases with higher electricity generation. Generally, a power plant with a capacity factor of 100% produces electricity continuously. It can be seen that the monthly capacity factor varies between 21.62% for February (the lowest) and 22% for July (the highest), with an annual value of 21.81%. This implies that the capacity factor for each month is very close. This annual capacity factor falls in the 20 to 40% range, which is usually recommended for a typical wind power plant [40]. Furthermore, the annual capacity factor is nearly comparable to the global average capacity factor of about 39% for the onshore utility-scale wind power plants deployed in 2021 [31]. The global weighted average capacity factor for onshore wind grew in 2021 due to increased deployment in countries with excellent wind resources [31].

It is seen that the 10 MW wind power plant generates about 19.103 GWh of electricity annually. According to the Energy Commission [20], Ghana's annual electricity consumption per capita is about 586 kWh in 2021. In view of this, installing the 10 MW wind power plant could supply electricity to about 32,599 people. Moreover, this is equivalent to

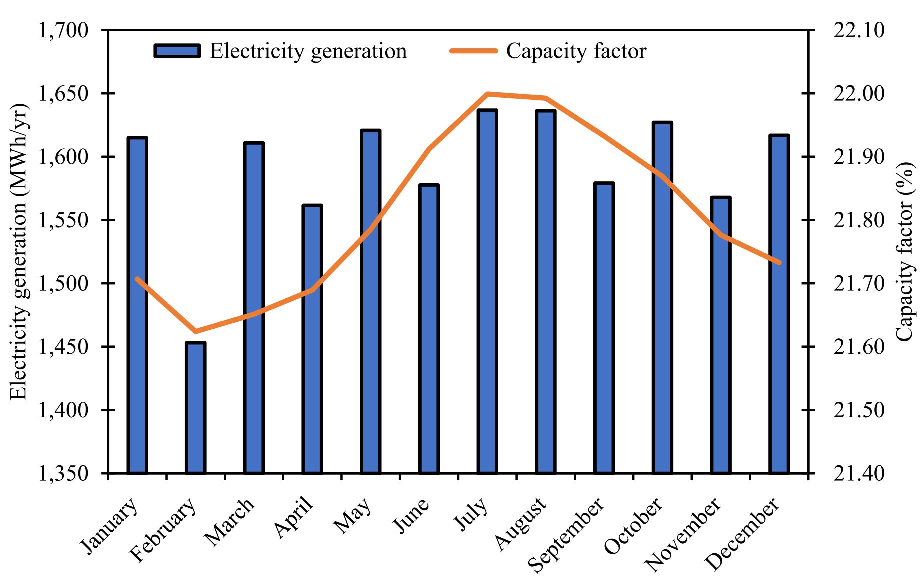

Fig. 4. Wind power plant electricity generation and capacity factor.

Table 8
Similar studies on LCOE for onshore wind power plants.

Authors	Study Location	Capacity MW	LCOE (US$/kWh)
Niyomtham et al. [48]	Central region, Thailand	15	0.093
Adeyeye et al. [2]	Moundou, Chad	1.5	0.250
	Ruhengeri, Rwanda	1.5	0.520
	Fez, Morocco	1.5	0.150
	Nakuru, Kenya	1.5	0.180
	Ilorin, Nigeria	1.5	0.270
Adnan et al. [3]	Umerkot, Pakistan	2.5	0.074
Okakwu et al. [54]	Port Harcourt, Nigeria	0.9	0.170
Pourasl & Khojastehnezhad [55]	Sevcenko, Kazakhstan	3	0.113
Shaahid et al. [68]	Khamis-Mushayt, Saudi Arabia	15	0.102
Charrouf et al. [12]	Illizi, Algeria	1	0.088–0.114

providing electricity to about 8,150 households using the recent household size of 4 people [22]. Accordingly, the electricity supplied to the grid could meet Ghanaian households' energy needs. As of 2021, only 86.3 % of the national households had access to electricity [20] out of 7.3 million households [22]. Based on this current data, it can be deduced that using the 10 MW wind power plant for electricity generation could increase the household's electricity access rate by about 0.1%. Furthermore, the peak electricity consumed by residential customers in Ghana was 8,484 GWh in 2021. Hence, the proposed wind power plant could meet only 0.23% of Ghanaians' residential electricity needs in 2021.

3.2.2. Economic analysis

Table 1 presents the base case input parameters applied to determine the net present value (NPV), levelized cost of electricity (LCOE), simple payback period (SPB), annual lifecycle savings (ALCS), and equity payback. Table 7 reveals that the project's NPV is about $3.80 million, suggesting it is financially feasible. In addition, the annual lifecycle savings is about US$ 447,215.17. The simple and equity payback periods are 5.2 and 5.6 years, respectively. The estimated equity payback and simple payback are nearly comparable to the ones obtained in other studies [33,59,74,80]. Equity payback is the time the project developer takes to recover its equity from project cash flows. The equity payback considers project cash flows from the start and debt level, giving it a more accurate prediction of project value than the simple payback.

It should be noted that a feed-in tariff (FiT) of US$0.166/kWh was applied. Thus, the project becomes financially feasible and acceptable when the LCOE is lower than the FiT. The LCOE of an onshore wind farm is defined by its capital cost, capacity factor, O&M costs, and economic lifespan. Nevertheless, while all these components are significant in calculating a project's LCOE, some have a more significant impact than others. For example, the turbine's cost (including the towers) is the most critical portion of the overall installed cost of an onshore wind generating plant. Table 7 shows that the wind power plant LCOE is US$0.143/kWh, less than the FiT of US$0.166/kWh. The estimated LCOE is just 11% higher than the average Ghanaian electricity tariff (US$0.129/kWh) in 2021 [20]. This suggests that the LCOE from the 10 MW wind power plant may not be as competitive with the utility grid. The reason is that Ghana's electricity generation mix comprises fossil fuels (mainly thermal plants) and renewables (solar photovoltaic and biogas plants). Besides, the electricity prices in Ghana are heavily cross-subsidised, especially for residential customers [10]. Nevertheless, as the cost of wind power continues to decrease and the technology continues to improve, it is likely that wind power will become an increasingly important component of the overall energy mix and can help reduce the negative environmental and social impacts associated with traditional energy sources. Furthermore, this study's LCOE is comparable to the LCOE obtained in similar studies highlighted in Table 8. For instance, the LCOE is lower than the LCOE obtained for wind power plants in Chad, Rwanda, Morocco, Nigeria, and Kenya [2]. Correspondingly, the LCOE is just 34% higher than the LCOE obtained in Saudi Arabia [68]. The difference in LCOE might be due to installed capacity and the wind speed used for site assessment. Consequently, the LCOE is anticipated to decline as the scale of the plant's capacity increases because of economies of scale. On the other hand, a recent survey by IRENA [31] has revealed that the global weighted LCOE of new utility-onshore wind power projects is about US$0.050/kWh, which is about 96% lower than this study's LCOE. It is worth mentioning that comparison of LCOE to other studies is vital for identifying trends and patterns in the cost of wind energy, demonstrating the cost-competitiveness of wind energy compared to other energy sources, and identifying areas for improvement in wind energy technology and development.

3.2.3. Emission analysis

Wind turbines only emit zero emissions during energy generation. However, significant emissions are produced during wind turbines' manufacture, construction, maintenance, and decommissioning. Ghana's fuel for grid electricity comprises diesel, hydropower, solar, and natural gas [20]. Therefore, the model computed the grid emission factor by using the share of each fuel for electricity generation at a 4.5% transmission and distribution loss. Similarly, an onshore wind turbine GHG emission factor of 0.082 $kgCO_2/MJ$ [78] was applied to estimate the amount of CO_2 emitted per unit of electricity generated and supplied to the national electricity grid. The grid and wind turbine emission factors are $tCO_2$0.476/MWh and $tCO_2$0.2952/MWh, respectively. The results in Table 7 show that electricity exported to the grid emits about 6,048 tonnes of CO_2 annually. Therefore, based on the grid emission factor and electricity available for grid export, the 10 MW wind power plant could reduce about 3,038 tonnes of CO_2 annually. This is equivalent to saving about 33% of CO_2 every year. In view of this, the emission reduction is equivalent to approximately 1,305,487 L of gasoline not consumed, 7,065 barrels of crude oil not consumed, and 3,038 people reducing their energy use by 20%.

The potential reduction in CO_2 emissions from wind power is significant for Ghana, given its commitments to reducing GHG emissions under the Paris Agreement. In light of this, the country has committed to decreasing GHG emissions by 64 $MtCO_2e$, generating employment opportunities for more than one million individuals, and preventing approximately 2,900 fatalities by enhancing air quality [44]. Wind power can play a significant role in helping Ghana achieve these targets. By reducing the consumption of fossil fuels, wind power can help mitigate the negative impacts associated with their extraction, transport, and combustion. These impacts include air pollution, water pollution, land use, and biodiversity loss. In addition to the environmental benefits, Ghana's adoption of wind power can contribute to economic and social development. As mentioned earlier, wind power projects could create job opportunities in installation, operation and maintenance, supporting the country's sustainable economic growth. According to Aldieri et al. [4], wind power plant average direct and indirect job creation is about 11 jobs/MW, and 3 jobs/MW for operation and maintenance. This implies that the proposed 10 MW could create about 110 direct and indirect jobs and 30 jobs for the operation and maintenance sections. Wind power could also improve access to electricity in rural areas, where the grid network is less developed, and contribute to poverty reduction efforts. However, it is essential to note that wind power adoption in Ghana may face challenges. One of the primary challenges is the high upfront costs associated with wind power projects, which could be a barrier to investment in the sector. Additionally, the need for a stable and reliable grid network to accommodate the intermittent nature of wind power generation is a significant challenge that needs to be addressed. The analysis also highlights the importance of engaging with local communities to address potential social and environmental concerns related to wind power development.

Table 9
Key input parameters for sensitivity test.

Input parameter	Unit	Sensitivity ranges
Feed-in tariff	US$/kWh	0.13, 0.14, 0.15, 0.166[a], 0.18, 0.19
Capital subsidy	%	0[a], 20, 40, 60, 80, 100
Discount rate	%	6, 8, 10[a], 12, 14, 18
Inflation	%	6, 8, 10[a], 12, 14, 16, 18
Wind speed	m/s	3.3, 4.3, 5.3[a], 6.3, 7.3, 8.3

[a] Base case scenario values.

3.2.4. *Sensitivity analysis*

The sensitivity test shows the robustness and resilience of the proposed wind power plant. The goal is to ascertain the project's viability when specific salient input parameters change. The salient parameters for the sensitive test are given in Table 9. The impact of the feed-in tariff (FiT) on financial indicators has been investigated, as shown in Fig. 5 and Fig. 6. Generally, the FiT is a support system that provides remuneration for clean energy producers. The FiT would secure a guaranteed price for electricity generated by the private sector from wind energy sources, ensuring that independent power producers get compensated at a rate greater than their production costs. Usually, the lifetime of this support ranges from 10 to 20 years [26]. FiT is phased out over time or when renewables reach a particular share of energy generation. The approved FiT for utility-scale wind power projects in Ghana is approximately US$ 0.166/kWh [58]. In this study, FiT's period is assumed to be 20 years. Fig. 4 shows the impact of FiT on NPV and the annual lifecycle. It can be seen that NPV and annual lifecycle savings significantly increase as FiT increases. Thus, the project becomes a viable and attractive investment as FiT increases. Furthermore, the project has a negative NPV for FiT that is less than US$ 0.15/kWh. The implication is that the project is unattractive for investment since the costs (capital, operating, and maintenance) exceed the benefits (revenues from selling electricity into the grid). Therefore, stakeholders, decision-makers, and investors could substantially lose money after investing. The impact of the FiT on equity payback and simple payback is shown in Fig. 5. The results show a positive, parallel linear relationship between equity payback and simple payback as FiT increases. The positive, simple payback indicates that the project's annual costs incurred are lower than the benefits. Practically, simple payback and low equity payback are preferred when assessing the performance of renewable energy projects. It is seen that equity payback is lower than simple payback as FiT varies. This result demonstrates that increasing the FiT for the wind project could facilitate wind power developers recouping their initial investment (equity) out of the project cash flows generated. The same trend follows for simple payback time.

A financial incentive is a non-refundable contribution, grant, or subsidy that covers the project's upfront costs. Table 10 presents the impact of the capital subsidy on the project's economic indicators. The 0% is the base scenario when the wind power plant is deployed without incentives. It can be observed in Table 10 that capital subsidies have a positive influence on economic indicators. For example, the NPV and annual lifecycle savings increase significantly at each subsidy level, making the project more profitable and attractive for investment. On the other hand, the LCOE shows a substantial and drastic reduction as subsidies increase. A key observation from Table 10 reveals that at the 100% subsidy level, the LCOE becomes competitive with the global weighted LCOE of new utility-scale onshore wind power projects. A similar trend exists for simple and equity payback, which also reduces with increasing subsidy levels. This implies that higher subsidies would cause a developer to recoup its investment in the shortest time possible. These results demonstrate that the proposed 10 MW wind power plant is sensitive to variations in capital subsidies. Therefore, obtaining capital subsidies on the upfront cost could make the project more profitable and attractive to stakeholders, decision-makers, and developers.

Most emerging nations, including Ghana, have weak currencies due to inflation and other economic indicators. Also, due to the currency rate volatility and its implications on items, including plant machinery and equipment, it is necessary to study the impact of inflation and discount rate variations on the project. Fig. 7 shows the effect of discount and inflation rates on NPV and LCOE. By gradually varying the inflation rate

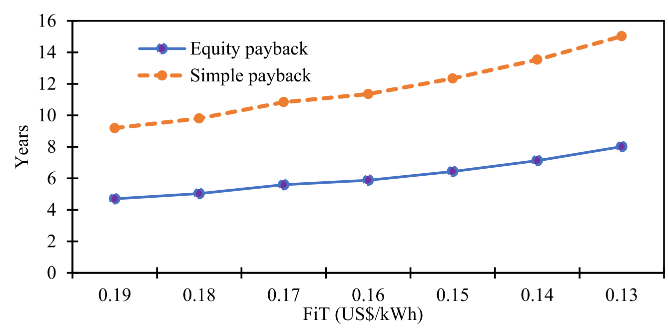

Fig. 6. Impact of FiT on equity payback and simple payback.

Table 10
Impact of financial incentives on key economic outputs.

Capital subsidy (%)	NPV (million US$)	LCOE (US $/kWh)	Simple payback (Years)	Equity payback (Years)	Annual lifecycle savings (US $/yr)
0	3.81	0.143	5.20	5.60	0.45
20	6.67	0.125	4.19	4.40	0.78
40	9.52	0.107	3.10	3.30	1.12
60	12.38	0.090	2.10	2.16	1.45
80	15.24	0.072	1.05	1.07	1.79
100	18.10	0.055	Immediate	Immediate	2.13

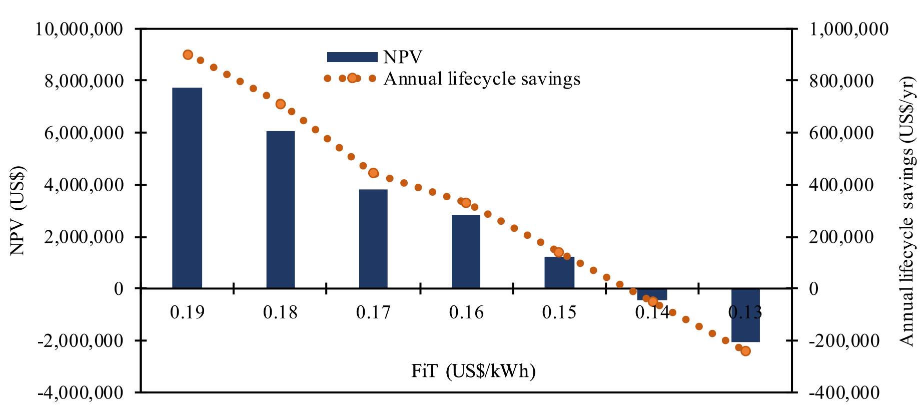

Fig. 5. Impact of FiT on NPV and annual lifecycle savings.

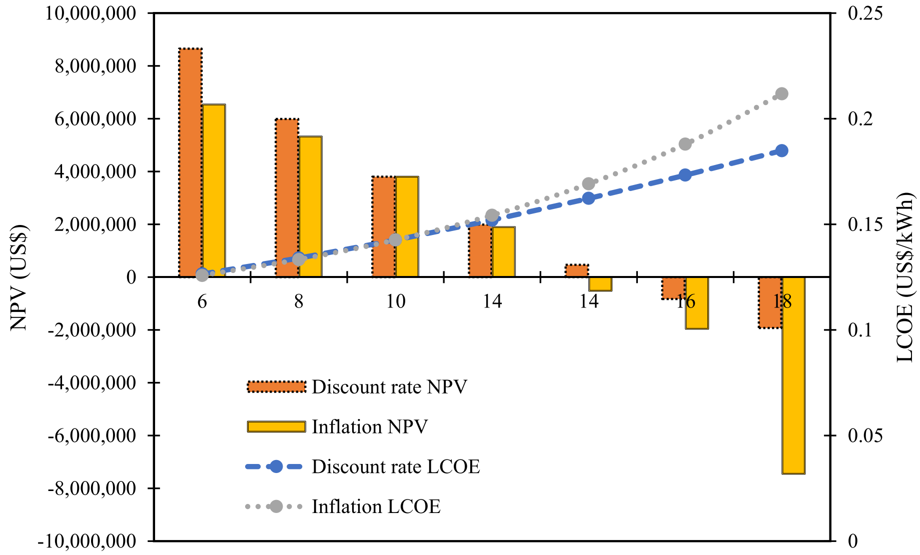

Fig. 7. Impact of discount and inflation rates on NPV and LCOE.

Table 11
Variations in wind speed at the study site.

Wind Speed (m/s)	Electricity (GWh/yr)	NPV (million US$)	LCOE (US $/kWh)	Simple payback (Years)	Equity payback (Years)	Annual lifecycle savings (million US$/yr)	Capacity factor (%)
3.3	5.50	−15.42	0.495	30.53	0	−1.81	6.3
4.3	11.78	−6.54	0.231	9.46	0.54	−0.77	13.5
5.3	**19.10**	**3.81**	**0.143**	**5.24**	**5.60**	**0.45**	**21.8**
6.3	26.11	13.72	0.104	3.67	3.79	1.61	29.8
7.3	31.53	21.36	0.086	3.00	3.00	2.51	36.0
8.3	35.06	26.36	0.078	2.70	2.70	3.10	40.0

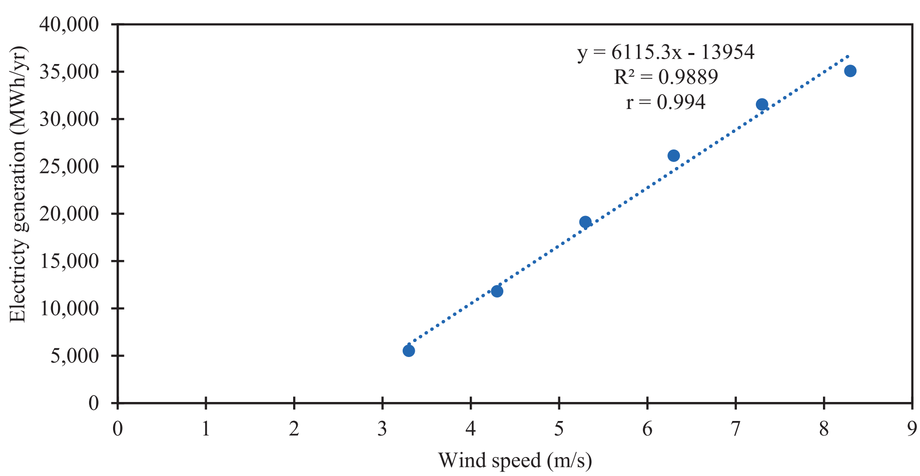

Fig. 8. Correlation between annual electricity generation and variations in wind speed.

from 6 % to 18 % for the discount rate values of 6 % to 18 %, it can be readily seen that the NPV decreases proportionally with increasing inflation and discount rates. Nevertheless, lower discount and inflation rates tend to reduce the LCOE. The results show that a discount and an inflation rate above 14% would make the project unattractive for investment. Hence, local and international banks could give lower discount rates on money borrowed for such a project to make it financially viable for investment.

Table 11 presents wind speed's impact on the study site's technical and economic indicators. It can be noted that electricity generation increases as wind speed increases. This tends to increase the capacity factor of the wind power plant since the power output increases as a cubic function of wind speed. Similarly, the NPV increases as the wind speed increases. Conversely, the LCOE decreases significantly at higher wind speeds. This is due to the higher electricity generated by, the higher wind speed. As seen, the payback period is shorter when the wind speed increases. The correlation between annual electricity generation and variations in wind speed is shown in Fig. 8. It can be seen that electricity generation follows the same trend as wind speed. The results show a positive linear relationship (with a correlation coefficient of 0.994) between electricity generation and variations in wind speed. The implication is that there is a high correlation between electricity

Table A1
Decision matrix for MCDM methods.

Turbine name	Ratedcapacity (MW) [a]	Capacity factor (%)[b]	Gross energy production (MWh/yr) [b]	Specific yield (kWh/ m^2)[b]	Energy for grid export (MWh/yr)[b]	Hub height (m) [a]	Rotor diameter (m) [a]	Swept area (m^2) [a]	Cut-in speed (m/s) [a]	Rated wind speed(m/s) [a]	Capital cost (10^6 UDS)[c]	Carbon dioxideEmissions (tCO2/yr)[b]	Operating and maintenance cost (10^3 US$) [c]
SL1500/70–65 m	1.5	24	3,678.97	810.30	3,154.51	65	70.4	3,893	3	12	2.11	38.24	66.75
SL3000/ 100–110 m	3	30	9,205.64	981.51	7,893.33	110	101.19	8,042	3	13	4.23	95.68	133.50
AW-70/1500 Class I – 80 m	1.5	25.30	3,876.60	863.72	3,323.97	80	70	3,848.46	4	14	2.11	40.29	66.75
DEWIND 62–68.5 m	1	27.39	2,798.29	794.74	2,399.38	68.5	62	3,019.07	3	15	1.41	29.08	44.50
ENERCON – 82 E2 2 MW – 98 m	2	30.18	6,167.61	1001.39	5,288.38	98	82	5,281.02	3	13	2.82	64.10	89.00
NORDEX S77 – 90 m	1.5	28.43	4,357.34	802.34	3,736.18	90	77	4,656.63	4	13	2.11	45.29	66.75
G80-2 MW – 60 m	2	23.27	4,753.89	810.93	4,076.20	60	80	5,026.55	3	17	2.82	49.41	89.00
G80 RCC – 78 m	1.8	26.46	4,865.00	829.89	4,171.47	78	80	5,026.55	4	16	2.54	50.56	80.10
VESTAS V90-3.0 MW – 90 m	3	23.91	7,327.02	987.55	6,282.52	90	90	6,361.73	4	15	4.23	76.15	133.50
VESTAS V90-1.8 MW – 80 m	1.8	30.67	5,639.48	760.10	4,835.54	80	90.3	6,361.73	4	13	2.54	58.61	80.10
LTW90-2 MW	2	34.35	6,797.47	939.80	6,018.47	97.5	90.3	6,404	3	13	2.82	68.24	89
LTW90 1 MW	1	21.30	4,600.62	578.72	3,706.14	90	90.3	6,404	3	9	1.41	42.02	44.50
A-1000/S – 70	1	21.81	2,227.90	834.19	1,910.30	70	54	2,290	4	15	1.41	23.15	44.50
A-1500–80	1.5	28.51	4,368.55	804.34	3,745.79	80	77	4,657	4	12	2.11	45.40	66.75
FL2500/90–85 m	2.5	25.97	6,633.39	894.02	5,687.77	85	90	6,362	4	15	3.52	68.94	111.25
FL3000 – 100 m	3	34.97	10,716.86	804.44	9,189.12	100	120.6	11,423	3	13	4.23	111.38	133.50
HSW 1,000/ 57–55 m	1.05	20.69	2,218.92	745.60	1,902.60	55	57	2,551.76	4	14	1.48	23.06	46.73
W2E-100/ 2.5–85 – 100 m	2.5	30.07	7,679.48	838.39	6,584.74	85	100	7,854.0	4	12	3.52	79.81	111.25
WWD-3 - D90 – 100 m	3	26.14	8,011.05	1079.74	6,869.04	100	90	6,361.74	4	13	4.23	83.26	133.50
ECO 122–139 m	2.7	35.63	10,460.19	720.89	8,426.47	89	122	11,689.0	3	13	3.80	95.54	120.15

[a] Data retrieved from original equipment manufacturers datasheet.
[b] Output extracted from RETScreen.
[c] Cost obtained from similar projects in literature.

generation and wind speed. These results demonstrate that higher wind speeds at the study site will likely make the wind power plant very competitive for investment. Furthermore, the findings from the sensitivity test could be used to measure the viability of wind power plants at other sites in Ghana with similar wind speeds to those investigated in this sensitivity test.

4. Conclusion

This study has assessed wind turbines suitable for developing wind power plants in Adafoah using multicriteria decision-making methods. Accordingly, a techno-economic and GHG emission analysis using the optimal wind turbine to develop a 10 MW utility-scale wind power plant has also been assessed. The findings revealed that wind turbines ranging from 1 MW to 1.5 MW are suitable for the study site. Also, the results indicate that installing the 10 MW wind power plant at Adafoah would generate about 19.10 GWh of electricity annually, at a capacity factor of 22%. The annual electricity generation can meet the electricity needs of about 32,599 people. This is equivalent to providing electricity to about 8,150 households using the recent household size of 4 people in Ghana. Also, this is equivalent to increasing Ghanaian households' electricity access rate by about 0.1%. Deploying the 10 MW wind power plant saves 3,038 tonnes of CO_2 annually compared to the utility grid. Additionally, the wind power plant LCOE is approximately US$0.143/kWh, less than the FiT of US$0.166/kWh. The sensitivity analysis also demonstrates that the project's financial indicators are resilient to capital subsidies, inflation, discount rate, FiT, and wind speed variations. The study's findings indicate that wind energy can significantly meet Ghana's electricity demand and improve access to electricity. Access to electricity is critical for economic and social development. Hence, installing the wind power plant could help improve the standard of living for the people in Adafoah and its surrounding areas. Similarly, developing the wind power plant could reduce Ghana's reliance on fossil fuels and promote sustainable development. This is particularly important given the growing concern about climate change and the need to reduce GHG emissions. The successful implementation of this project could serve as an example for similar projects in Ghana and other African countries, providing a sustainable and climate-friendly development model. The study suggests that wind energy investment can be more financially viable with state support and infrastructure development, such as government support for capacity building, which can enhance Ghana's wind power development. Future studies should focus on the micro-siting of wind farms (wind farm layout and optimal way of arranging the wind turbines) and modelling the wind turbine wake at the study site to optimise wind power generation in the region.

CRediT authorship contribution statement

Flavio Odoi-Yorke: Conceptualization, Methodology, Software, Writing – original draft, Writing – review & editing. **Theophilus Frimpong Adu:** Methodology, Data curation, Writing – original draft, Writing – review & editing. **Benjamin Chris Ampimah:** Visualization, Investigation, Writing – review & editing. **Lawrence Atepor:** Supervision, Writing – review & editing.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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Appendix

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